

OUT-OF-CYCLE OPPORTUNITIES

I. GENERAL INFORMATION

A. Closing Date: 30 September 2023. This Opportunity Notice shall remain in effect for abstract submissions until the closing date has passed.

B. TTA-Specific Topics: If listed under a TTA, topics are grouped and numbered in the following sections by TTA per numbered domains.

C. Out-of-Cycle ‘Other Test Technologies’ Topic

This category encourages offeror consideration to address technology research opportunities (which may include studies) in areas that are not identified among currently active/ listed topics but may be beneficial and critical to the future of the technologies that address DoD test needs for rapid transition into the T&E infrastructure. This may identify potentially revolutionary weapon and sensor systems needing T&E capabilities that are currently “unknown unknowns” to the Government.

All Out-of-Cycle topics first require an abstract submission prior to a white paper submission per **Enclosure 1 – Research Opportunity Guidance**.

D. Table Legend

In-Cycle vs. Out-of-Cycle Opportunity Notice Type

TTAs: Test Technology Areas’ initialisms are listed below:

- Autonomy and Artificial Intelligence Test (AAIT)
- Advanced Instrumentation Systems Technology (AIST)
- Command, Control, Communications, Computers, Intelligence, and Software Intensive Systems Test (C4T)
- Cyberspace Test Technology (CTT)
- Directed Energy Test (DET)
- Electronic Warfare Test (EWT)
- High Speed Systems Test (HSST)
- Space Systems Test (SST)
- Spectrum Efficient Technology (SET)

‘Part.’: Indicates if the TTA is participating in this opportunity release (Yes/No)

TTA Topics: Indicates if TTA-specific topics are listed herein (Yes/No)

‘Other’: Indicates if the ‘Other Test Technologies’ topic applies to this release (Yes/No).

OUT-OF-CYCLE OPPORTUNITIES**E. Tabular Opportunities Listings**

The tables below show TTA participation for the listed ‘cycle’ type. Only TTA domains where topics are issued will be numbered in this Opportunity Notice. Specified topics within this document can be accessed easily using the hyperlinks in the tables below:

Out-of-Cycle

TTA	Part.	Topics	‘Other’
AAIT	Yes	Yes	No
AIST	Yes	Yes	No
C4T	Yes	Yes	No
CTI	Yes	Yes	No
DET	No	No	No
EWT	No	No	No
HSST	Yes	Yes	No
SST	No	No	No
SET	Yes	Yes	No

TOPICS: AAIT - AUTONOMY AND ARTIFICIAL INTELLIGENCE TEST[Return to Table](#)

Domain 1 – Autonomous System Performance Assessment
(ASPA)

1. Assess Effectiveness of Human/System Interactions

As unmanned, autonomous, and artificial-intelligence-based systems become more common, it is important for users of these systems to interact effectively with them, to understand how they operate, and understand how they are controlled.

The interaction between these systems and the humans who are in related roles is complex. Humans need to be able to interact with these systems in many ways: to give these systems both commands and status of mission activities, and to gather information from these systems for awareness of mission activities and status. In short, to collaborate, cooperate, and control. To ensure that the human-machine interactions are effective, testers need novel technologies to evaluate exactly how effective the human-machine interaction is.

To ensure that the DoD delivers effective autonomy for use on the battlefield and to enable deployment of effective human-machine teams, testers need tools and technologies, testing standards, and necessary frameworks to quantitatively evaluate the level of effectiveness of human/system interactions.

2. Assess Trust in Use of Autonomous and Artificial Intelligence-Based Systems

As unmanned, autonomous, and artificial-intelligence-based systems become more common, it is important for users of these systems to be able to trust them to perform their missions effectively and safely.

Current unmanned, autonomous, and artificial-intelligence-based systems do not incorporate built-in quantifiable metrics of operational trust. Metrics defining operational trust are not well understood, beyond subjective measures and links to risk induced by vulnerability associated with delegating authority to such systems. Additionally, it will be key to incorporate the emerging understanding of mission autonomy and how to measure trust in it.

In addition to the systems themselves, there is a lack of existing frameworks within the DoD to define quantifiable metrics for trust and a lack of uniform concepts for architecting such trust. Additionally, the testing, evaluation, verification, and validation (TEVV) of autonomous systems technology needs further improvement to provide metrics for trust. Embedding operational trust into autonomous systems design and mission capability is a challenge. Testers need innovative technologies to define metrics, observe measurable behaviors or processes, continuously monitor, assess performance in real-time, track and audit quantifiable metrics, and quantify operational trust.

To ensure that the DoD delivers “trusted autonomy” on the battlefield, testers need tools and technologies, testing standards, and necessary frameworks to quantitatively evaluate the level of trust in human/system interactions.

TOPICS: AAIT - AUTONOMY AND ARTIFICIAL INTELLIGENCE TEST

Domain 2 – Counter-Unmanned Aerial Systems (C-UAS)

1. Counter-Unmanned System (C-UxS) Performance Assessment

As unmanned, autonomous, and artificial-intelligence-based systems become more common, countermeasures for those systems, in all domains (Unmanned Air, Surface, Subsurface, Ground = UxS), will become more common. Testers need the ability to test the performance of blue Counter-UxS systems in safe, controllable environments in advance of deployment to live operations (and live test).

Testers need to understand C-UxS performance and identify risk areas in the safety of a modeled environment, so that subsequent live testing may be performed safely and efficiently. C-UxS can take many forms. The systems can be electronic-warfare (EW) based, directed-energy (DE) based, kinetic, etc. The risks associated with live testing of blue C-UxS are based on the kind of countermeasure. As a step toward readiness for live testing, we must prepare to do so by predicting countermeasure system performance in a modelling and simulation (M&S) environment. This drives the need to perform as much testing as possible in modeled environments. In M&S, high-risk or unexpected behaviors can be identified and tested safely, minimizing problems that would otherwise have been found during live test.

Testers need tools to evaluate autonomous blue C-UxS systems in an M&S environment. There is a lack of M&S environments and models for EW-based, DE-based, kinetic, and other C-UxS systems. A C-UxS-capable M&S environment is needed, along with models and behaviors for the countermeasure techniques and the vehicles necessary to populate a realistic scenario.

Testers need tools to develop elements of the M&S environment with sufficient fidelity to assess performance: blue C-UxS systems, vehicles, sensors, and other required elements of a C-UxS mission.

TOPICS: AIST - ADVANCED INSTRUMENTATION SYSTEMS TECHNOLOGY[Return to Table](#)

Domain 1 – Advanced Sensors (AS)

1. Pressure-Sensitive Paint (PSP) Camera Sensor Systems

PSP uses a special paint applied to conventional test articles. The paint fluoresces red light with an intensity inversely proportional to the oxygen partial pressure at the surface when excited by near ultra-violet light, i.e., low-pressure regions are bright while high-pressure regions are dim. PSP is used in wind tunnel testing to measure surface pressure across the surface of test articles and, most importantly, when integrated can provide distributed loads required for structural analysis of new hypersonic weapon systems. Camera technology is key for making PSP measurements.

Unfortunately, the cameras (and sensors) are no longer being manufactured with the shift to complementary metal-oxide semiconductor (CMOS) sensors. Next-generation PSP sensor development, followed by camera integration, is required.

TOPICS: AIST - ADVANCED INSTRUMENTATION SYSTEMS TECHNOLOGY

Domain 2 – Human Performance Measurement and Assessment
(HPMA)

1. Real Time Casualty Assessment (RTCA)

A capability to assess and track casualty assessments during test events is needed, especially for attacks from and attacks against unmanned systems for both kinetic and non-kinetic weapon (such as cyberattacks). The representations, trajectories, and impacts of kinetic weapon should not be limited by unrealistic constraints (such as, a laser being unable to penetrate a cloth curtain but a bullet can.)

TOPICS: AIST - ADVANCED INSTRUMENTATION SYSTEMS TECHNOLOGY

Domain 3 – Range Environmental Encroachment (REE)

1. Soft Catch System for Large Caliber Munitions

Soft catch capability is needed to assess munitions internal ballistics, strength of design, and function of critical components, where the munition can be fired at operational velocities, and captured in a way that does not damage the projectile or the environment. There is a need to perform a soft catch operation of large caliber munitions (75mm and above) fired at operationally representative velocities in order to assess the behavior of projectile features and components (e.g., guidance electronics, stabilizing fins) during firing as part of the RDT&E open air range test phases.

Current practices have limited success, and a significant portion of test munitions are damaged during recovery, which requires additional testing and prototype munitions. This increases the technical risk to systems under test, due to increased cost and timelines. It is unknown at this time if the larger and heavier long range artillery munitions will be compatible with current practices due to their vastly higher energy, and whether they may also be limited by range boundaries or environmental concerns.

TOPICS: C4T – COMMAND, CONTROL, COMMUNICATIONS, COMPUTERS, AND INTELLIGENCE (C4I) AND SOFTWARE INTENSIVE SYSTEMS TEST[Return to Table](#)

Domain 1 – Command and Control (C2) Analysis in Multi-Domain Environments (C2MDE)

1. AI/ML to Improve Automated C4I Test Planning, Design, and Assessments

Generally, within a multi-domain operations (MDO) environment, the whole of the system-of-systems (SoS) performance is greater than the sum of the individual system's performance. Current effectiveness evaluation capabilities among Services are fragmented and are specific to the systems executing within one Service's mission context (Army analysis capabilities are limited to Army systems, Navy analysis capabilities are limited to Navy systems, etc.) When a SoS is composed of systems from multiple Services operating across multiple domains, manual processes are required to analyze cross-Service DoD capabilities in a mission context. Technologies are needed to plan, set up, and design MDO tests incorporating a modeling component to improve on current Design of Experiments process through artificial intelligence/machine learning (AI/ML) to minimize test design and planning times from current months to days. These technologies could also be extended to automate the assessment of the mission context involving effectiveness, timeliness, completeness, and accuracy measures from all systems impacting a specific mission thread.

Additionally, efforts in this area will explore technologies to identify key enabling events in the mission thread to provide automated semantic and transformational analysis across the test scenario with varying participating systems. Transformational analysis is defined as data flowing into one interface of a system and out another interface and cascading through multiple systems within the SoS. Technologies are needed to drill down to individual system performance and contribution, and then determine the impact to the effectiveness of the SoS performance.

Current test planning and test scenario development activities involve manual and labor-intensive processes. Efforts in this area will explore technology development for automation of test planning and scenario development with AI/ML techniques to address these MDO planning needs:

- Technologies are required to automatically select, via an ontology and knowledge base, the appropriate live and simulated entities in support of the test strategy.
- Technologies should address multiple mission threads executed simultaneously across multiple domains that impact the C2 alternative actions.
- Technologies should address the ability to dynamically make near real-time changes to test planning to rapidly create new updated test architectures and test scenarios.
- Technologies should address creation of a learning environment to improve test plan/design updates based on model consumption of live test data or operational data collected at edge devices, to continually improve future planning events.

TOPICS: C4T – COMMAND, CONTROL, COMMUNICATIONS, COMPUTERS, AND INTELLIGENCE (C4I) AND SOFTWARE INTENSIVE SYSTEMS TEST

- AI/ML technologies should address the full range of military test and evaluation capability, to include rule bases, synthetic data, and computer vision to drive maturation of the proposed AI/ML technologies.

2. Intelligent Testing Technologies for C4I Warfighter Systems Operating within One Domain and Influencing Actions Across Other Domains

Warfighter systems are increasingly incorporating artificial intelligence/machine learning algorithms across many types of platforms, especially the command, control, communications, computers, and intelligence (C4I)/C2 planning and dynamic replanning mission capabilities. C4I warfighter systems/SoS that implement artificial intelligence/machine learning algorithms to perform operational C2 missions must be adequately assessed. A core set of intelligent testing technologies is needed to test these C4I systems at a system and SoS level within an operationally relevant, mission-oriented, live and simulated distributed environment. Intelligent testing technologies are required to assess the full C2 hierarchy operating across multiple domains of warfare.

Technologies are needed for:

- Testing of C2 hierarchy of C4I warfighter systems/SoS operating across the collective set of Service participants in multi-domain operations.
- Automated execution, analysis, and reporting to test C4I system/SoS.
- Assessing a C4I system/SoS that is a learning system, which may change and evolve throughout testing, and being able to characterize the performance in a repeatable manner.
- Explainable AI to focus on the results of the C4I system/SoS performance under standard and degraded operating conditions, as well as identify any resulting range safety factors.
- Computer vision and analysis of C4I systems/SoS operating across multiple warfighting domains.

TOPICS: CTT – CYBERSPACE TEST TECHNOLOGY[Return to Table](#)

Domain 1 – Cyber-Physical Systems (CPS)

1. Tools and Methodologies to Enable Automation of Testing for Components of Cyber-Physical Systems

The Department of Defense Test Resources Management Center (TRMC) is seeking innovative tools and methodologies to enable automation in the testing of software and hardware of the cyber portions of the components of Cyber-Physical Systems (CPS). Tools and methodologies for testing CPS in ships, aircrafts, ground vehicles, and other weapons systems are the higher priority, but tools to address Industrial Control Systems (ICS) and DoD critical infrastructure are also within this topic's need. The anticipated technology development should focus on tools that primarily will be used for in-situ testing, where the tool will be used to test CPS cyber components while they are installed in their normal operational or development environments. These tools could be used for testing a single component while it is installed in the system, or more comprehensive testing like cooperative vulnerability assessments and red team events. Additional use cases include testing on cyber ranges and in laboratories (e.g., system integration labs). Of particular interest are tools to test components that are connected via protocols other than IP.

Tools desired include: software-only tools for deployment on COTS or GOTS hardware with interfaces for the necessary busses/protocols, handheld tools with built-in interfaces for one or more CPS busses/protocols, and tools that run on a laptop or tablet with an external interface to connect to CPS busses. For software-only offers, the COTS or GOTS hardware required to host the tools must be stated.

Tools and methodologies developed within this research are expected to be integrated into a GOTS testing toolkit intended to be maintained by the Government or government contractors.

All tools should demonstrate the ability for both a manual and automated mode. Automation can be accomplished via external tools such as AAFT (Automated Attack Framework for Test & Evaluation) or fuzzers such as VMF (Vader Modular Fuzzer), or may be built into the tool itself.

Offer phasing strategy should consider phase lengths of up to a year each with a maximum of four. Each phase should conclude with a demonstration using a specified system under test (SUT). However, in Phase I the Government will not provide access to a SUT for development and testing, but may elect to independently verify operation of the delivered tool on a representative SUT. During subsequent phases, the Government will likely identify a government transition partner for the delivered tool and that entity may provide for testing and feedback at regular intervals.

Concept of Operations

As shown in the figure below, the Government's test event Concept of Operations (CONOPS) for these tools is expected to encompass three phases. In the pre-test phase, the SUT

TOPICS: CTT – CYBERSPACE TEST TECHNOLOGY

will be specified (preferably using hardware and software bill of materials - HWBOM and SBOM), the test will be designed, and appropriate test vectors will be selected. In this phase handheld tools will be connected to IT devices to enable the preparation for the test, while laptop-based tools will be connected directly to test infrastructure resources indicated by the blue cloud in the figure below. In the testing phase, the tool will be connected to the SUT via the appropriate bus (e.g., MIL-STD-1553) and test vectors will be executed, data recorded, and results collected. In the testing phase, the tool may be controlled by automation software such as AAFT or a fuzzer, or in the case of laptop-based tools, that automation software may reside on the laptop and control the tool via an API. In the post-test phase, data is uploaded from the tool (via an IT device if required) to the test infrastructure where the data can be analyzed, test reports created, and test results uploaded to appropriate data stores.

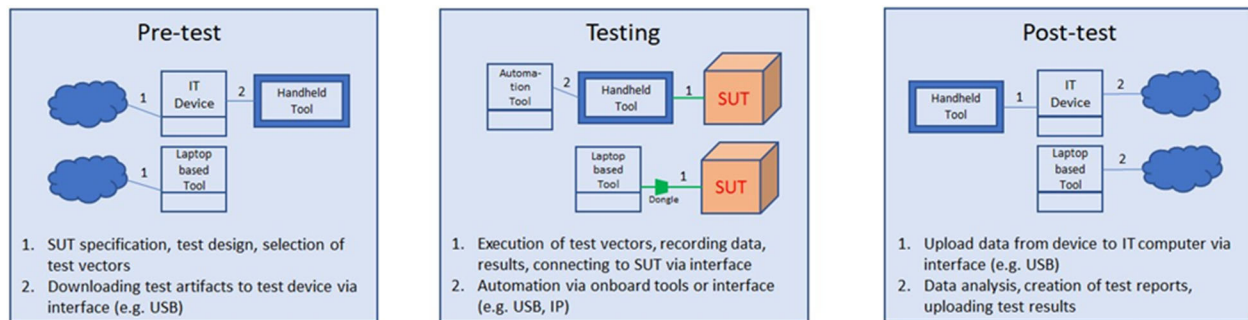


Figure 1: Three-Phase Testing

Background

TRMC is currently funding the development of two testing automation tools, Automated Attack Framework for Test & Evaluation (AAFT), and the Vader Modular Fuzzer (VMF). These tools provide various decision algorithms that may form the core for automated testing. It is desired, but not required that tools developed in this effort provide an API that will enable AAFT or VMF to use the tools.

Penn State University/Applied Research Lab is developing AAFT for TRMC to provide support for DoD Red Teams and the DoD testing community, to automate basic and intermediate cyber testing with a current focus of the automation of testing on traditional IT networks using IP protocols. AAFT contains a repository of attack scripts, which are atomic steps that may be taken in a testing sequence. Each attack script provides the executable to perform the atomic step, along with pre-conditions for the script, post-conditions for the script, and other metadata. AAFT works by creating and maintaining an attack tree for a given SUT using all scripts allowed by the testing rules of engagement. It has modes to automatically or manually select the next attack script to execute using various strategies, including strategies that mimic advanced persistent threats.

Draper Laboratory is developing VMF for TRMC. VMF is a framework to create a specific fuzzer with a selected fuzzing strategy using modules in the framework as specified by a configuration file. VMF includes an implementation of AFL++, the daughter of the american fuzzy lop (AFL) fuzzer, and also incorporates modules to implement other open-source fuzzing

TOPICS: CTT – CYBERSPACE TEST TECHNOLOGY

algorithms in various modules, including various execution engines, seed selection algorithms, mutation algorithms, bug oracles, and crash triage modules.

Problem Definition

Offers are sought for technology development that addresses the design and implementation of tools and methodologies to enable automation of testing for Cyber-Physical Systems. The technology development prototype should describe a single tool or a family of tools, including the protocols by which the tool will communicate with the SUT, the test vectors (attack scripts) the tool will be capable of executing, the mechanism to be used for automation, and the pre-test and post-test capabilities of the tool. Test vectors should be identified by the techniques enumerated in the MITRE ATT&CK and the ATT&CK for Industrial Control Systems (<https://attack.mitre.org/>.)

2. Tools and Methodologies to Create Virtual Machines of Components of Cyber-Physical Systems

The Department of Defense Test Resources Management Center (TRMC) is seeking innovative technologies to create Virtual Machines (VMs) of the cyber components of Cyber-Physical Systems (CPS), regardless of CPU, chipset, or other hardware considerations, for use in testing. These tools and methodologies should be generic in nature and enable the creation of VMs for similar classes of devices, vice creating a VM of a single target. As discussed below, two classes of virtual machines (high fidelity and medium fidelity) are desired.

While tools that are fully automated are eventually desired, it is anticipated that technology development efforts should produce tools within this problem space addressing at least a “90% solution” that can be refined and completed with human interaction. TRMC is also interested in methods to quantify or qualify the fidelity of the developed Virtual Machine.

Tools and methodologies developed in this research are expected to be integrated into a toolset for use in testing on cyber ranges and in dedicated cyber testing labs. The solution(s) are intended to be maintained by the Government or government contractors. Tools developed under this effort will need to be delivered as both executables and source code, with clear guidance on how to use the tools and when the tools are appropriate. The offeror’s submission in response to the Opportunity Notice should include within the research effort an adequate detailing of the methodology that a qualified cyber tester could follow to create a VM, along with a Rough Order of Magnitude (ROM) for the labor expenditure Level of Effort (LOE) to create an automated or semi-automated tool.

Tools and methodologies are desired to create two classes of virtual machines. The first is high-fidelity VMs that are virtual versions of a system under test (SUT). These VMs will be used in test scenarios where the VM should use binary executables that are essentially the same as the physical SUT; thus inputs will be processed in the same manner on both, and cyber test vectors will produce the same effects to both the VM and the physical SUT. One example of the use for these VMs is in fuzzing. The second class is medium-fidelity VMs. These VMs are designed to be used in testing other VMs which represent the SUT. These VMs provide part of the test universe for a SUT and will not be tested directly. These VMs require medium fidelity of the

TOPICS: CTT – CYBERSPACE TEST TECHNOLOGY

artifacts they produce on the busses and in the inputs and outputs they provide to interact with the SUT.

Background

Current DoD cyber testing for CPS is accomplished using physical versions of the SUT. Physical versions of these systems are often costly, limited in their availability, and subject to damage in testing. In addition, more cost-effective testing can often be accomplished using virtual SUTs. Finally, state of the art testing methodologies like fuzzing require virtual targets.

Concept of Operations

The test scenario Concept of Operations (CONOPS) for these tools is conceptually simple. A user of the tool will create a high-fidelity VM of the SUT. This may be an iterative process. The user will use the same or similar tools to create VMs for the other components required for the test universe. One possible configuration for this test is shown in Figure 1. In this example, the test consists of three high-fidelity VMs, two representing the SUT and one representing a VM in the universe. In addition, the universe also consists of eight other VMs of lower fidelity. For illustrative purposes, it is assumed that these systems are connected via a bus (e.g., MIL-STD-1553), which is implemented virtually in the hypervisors and with the protocol in an IP tunnel between hypervisors. Figure 1 also includes a physics model running in one or more virtual machine(s) that calculates physical values for sensors, and virtual sensors that receive these values via their hypervisor and provide inputs to the various virtual machines. Finally, there may be a traffic-generation tool running in a virtual machine to provide traffic on the virtual bus. The process for creating virtual machines may be iterative. In this example, all the components are in virtual machines. Other use cases might include some physical systems connected via a physical bus with an interface to the hypervisors.

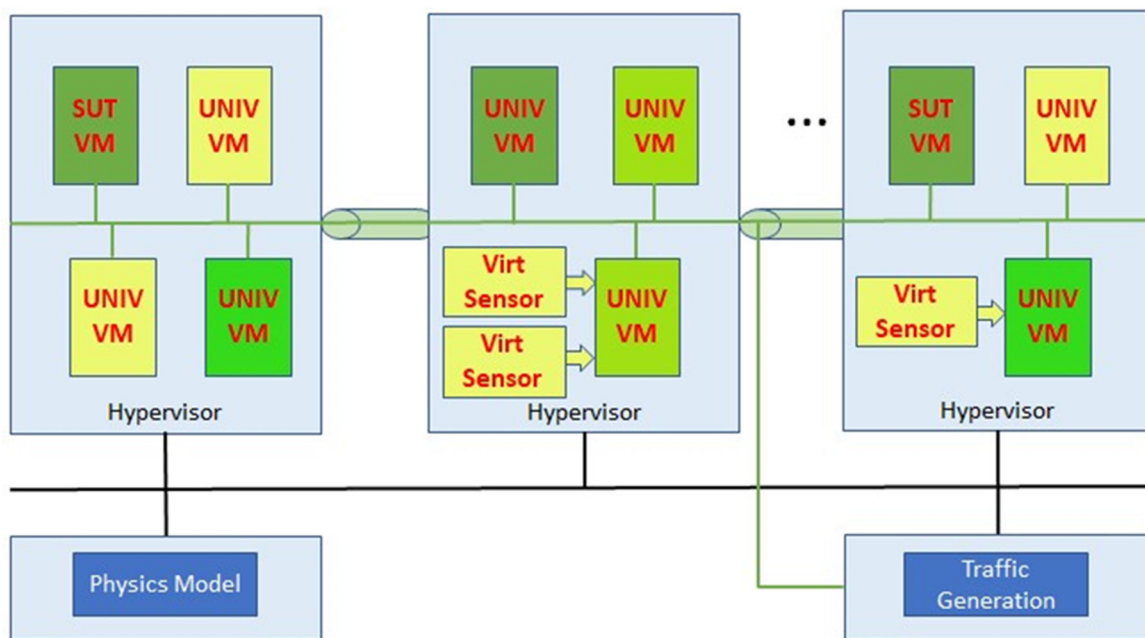


Figure 2: Possible Test Configuration

TOPICS: CTT – CYBERSPACE TEST TECHNOLOGY

Offers may assume some starting representation of the target including a) physical device, b) binaries for a physical device, c) source code and build instructions, d) description of the hardware, and e) description of the operating system. Any assumptions and starting representations must be clearly specified. If a hardware description is not provided or is incomplete, the tools should create a proposed HW layout. Offers should include a proposed methodology to estimate the fidelity of the virtual machine.

Problem Definition

The goal of this problem space is to develop tools that enable the creation of virtual machines for the cyber portion of cyber-physical systems. We have divided this problem into three technical areas. Offers may address one or more technical areas.

TA1 – High-Fidelity Virtual Machines

Innovative solutions are sought to create high fidelity VMs to enable testing of the cyber portion of a Cyber-Physical System. These tools should be generic in nature and enable the creation of VMs for similar classes of devices (e.g., line replaceable unit built upon a common core architecture), vice creating a VM of a single target. While tools that are fully automated are eventually desired, tools created under this problem space should constitute at least a “90% solution” that can be refined and completed with human interaction. Creation of both medium and high-fidelity virtual machines for the same system is allowable if an iterative approach is used.

For this problem space, high-fidelity VMs are defined as virtual versions of a system under test (SUT) that use binary executables that are essentially the same as the physical SUT; thus inputs will be processed in the same manner on both physical and virtual systems, and cyber test vectors will produce the same effects of both the VM and the physical SUT. Offers should include methods for creating virtual sensors for systems that receive sensor inputs. Virtual sensors should receive information on the simulated physical state of the CPS from a physics model and should provide an output in the form that will be consumed by a VM.

This technical area is restricted to tools that actually create a VM. Offers should include a set of target systems that will be used to demonstrate the capability of the tools. Offers may include one or more phases, each up to one year in duration, with clearly enumerated exit criteria for each phase. These criteria should include creation of VMs at a given level of fidelity leading up to high fidelity in the final phase, with a demonstration at the end of each phase. Solutions that use the QEMU hypervisor are preferred, but those using other hypervisors will be considered. Offers must include the hypervisor to be used and the method to be used to quantify or qualify the fidelity of the VM.

TA2 – Medium-Fidelity Virtual Machines

Innovative solutions are sought to create medium-fidelity VMs to enable testing the cyber portion of a complex Cyber-Physical System, which must be represented with multiple VMs. These tools should be generic in nature and enable the creation of VMs for similar classes of devices (e.g., line replaceable unit built upon a common core architecture), vice creating a VM of a single target. While tools that are fully automated are eventually desired, tools created under this problem space should constitute at least a “90% solution” that can be refined and completed with human interaction.

TOPICS: CTT – CYBERSPACE TEST TECHNOLOGY

For this problem space, medium-fidelity VMs are defined as virtual versions of a system or component that receive inputs and produce outputs on attached busses and protocols that are the same as those produced by the actual system. These VMs will be used as part of the test universe for a SUT and not tested directly. Internally, these VMs may be based upon different architectures and operating systems that the physical devices they represent. Offers should include methods for creating virtual sensors for systems that receive sensor inputs. Virtual sensors should receive information on the simulated physical state of the CPS from a physics model and should provide an output in the form that will be consumed by a VM.

Offers in this TA are restricted to tools that actually create medium-fidelity VMs; proposals should include a set of target systems that will be used to demonstrate the capability of the tools, and must include the hypervisor to be used and method to be used to quantify or qualify the fidelity of the VM. Offers may include one or more phases, each up to one year in duration, with clearly enumerated exit criteria for each phase.

TA3 - Virtual Busses

Innovative solutions are sought to create virtual versions of busses used in CPS. These virtual busses should run on QEMU hypervisors to allow for a VM running on the hypervisor to communicate on the virtual bus. They should also provide for interconnection between physical hypervisors (for example, tunneling the protocol over IP) to allow VMs hosted on one hypervisor to communicate with VMs on other hypervisors.

TOPICS: HSST – HIGH SPEED SYSTEMS TEST[Return to Table](#)

Domain 1 – Aerodynamics and Aerothermal (A)

1. Arc Heater Integrated Power Operations

Unlike current mid-pressure and high-pressure arc heater (MPAH) power supply operations, arc heater initiation with new thyristor-based power supplies used in the mid- to high-pressure range is not fully characterized. Low-pressure arc heater concept of operations (CONOPS) allows for successful operation of the arc-heater systems with thyristor-based power supplies; however, utilization of the same starting process with a thyristor-based power supply for mid- or high-pressure arc heaters is at a lower test readiness level. In order to reduce programmatic and technical risk to future facility power supply integration, a test methodologies project is desired to perform analysis, determine best practice control methodologies, and conduct modeling and computational simulations to gain information on starting a thyristor-based power supply for mid- to high-pressure arc heater systems. Additionally, to further reduce risk, the development and demonstration of a prototype test bed is desired to experimentally determine operational processes required to successfully initiate operation of a mid- to high-pressure arc heater while using a thyristor-based power supply.

Power limits of existing mid- to high-pressure arc heater systems are not well characterized, especially while utilizing higher power levels enabled by a thyristor-based power supply. Limits of the existing arc heater designs are estimated based on experimental information, but full characterization of these limits are not fully understood due to existing power supply limitations with operational facilities. Additionally, as power supply levels to the arc heater system increase, increased heat flux to the heater bore will be realized. If experienced, the increased heater bore heat flux has the potential to lead to higher levels of wall-arcing events. A higher number of wall-arcing events may lead to lowering the performance capability of new, future arc heater test assets, and/or reducing the availability of arc heater systems to support weapon system development programs. It is desired to initiate a demonstration study in coordination with the development of starting processes for mid- to high-pressure arc heater systems to fully characterize subatmospheric and atmospheric arc heater operational envelopes while utilizing a thyristor-based power supply. Knowledge gained on these physical limitations will assist in setting expectations for capabilities enabled with existing arc heater designs and will provide critical information to assist design of future, higher power, more stable arc heaters for weapon system development.

TOPICS: HSST – HIGH SPEED SYSTEMS TEST

Domain 2 – Flight Test and Evaluation (FTE)

1. In-Flight Sensing of Atmospheric Conditions by High-Speed Vehicles

On-board flight test hardware and software tools are needed to obtain, capture, store, and/or transmit in real-time the atmospheric air flight conditions that high-speed, hypersonic vehicles encounter along their flight paths. The measured atmospheric data must be converted into appropriate engineering units for real-time in-flight vehicle performance and operability assessments. Measurements of atmospheric air temperature, density, and velocity along the flight path ahead of or surrounding a high-speed flight vehicle are needed to support the developmental flight test and evaluation programs of the vehicles. Real-time atmospheric data are critically needed for setting or adjusting the vehicle aerodynamic attitude (pitch and yaw angles, control surface deflections, etc.) and propulsion system operation schedules (thrust, fuel flow, throttle settings, etc.) in flight. Accurate knowledge of the atmospheric conditions encountered by the flight test vehicle will allow improved correlation of the vehicle's aerodynamic, aerothermal, and propulsion performance to the atmosphere it will encounter over its flight test trajectory and is critical to establishing the vehicle's range and operability. In-flight air data sensing systems (ADS) and tools and technologies for prediction of atmospheric flight conditions on the flight paths of high speed, hypersonic vehicles are highly desired to better assess flight test variables and also to better understand flight vehicle performance as it accomplishes its test mission. ADS that minimally disturb or are non-intrusive to the freestream and/or boundary layer air flow over the vehicle are highly desired.

2. Broad Ocean Area Terminal Scoring Capabilities

The ability to score terminal impact of high-speed/hypersonic (HSH) weapon systems in the broad ocean area (BOA) is needed to support development of such systems. Traditionally, scoring information that is used to assess the performance of a weapon system's impact on the intended target is acquired via a land impact instrumented with the appropriate instrumentation. For flight tests that plan on a BOA impact, the ability to acquire similar scoring information is needed. This includes data such as impact location, debris field characterization, and impact angle and velocity. The ability to acquire such datasets at a similar fidelity and data quality as traditional land impacts will provide new opportunities to conduct flight tests that require terminal scoring since BOA impacts will be available. For a BOA impact, a terminal scoring capability that can be deployed shortly before the flight test is desired. This may involve the use of airborne assets to deploy the required instrumentation at the target location. A combination of sea-based and airborne sensors may be necessary to gather the required scoring information. Depending on how the instrumentation will be deployed, size, weight, and power considerations will need to be made based on what platform will be utilized for the deployment.

TOPICS: SET – SPECTRUM EFFICIENT TECHNOLOGY[Return to Table](#)

Domain 1 – Wireless Technologies (WT)

1. Enhanced AMT Capabilities for APSK

In order to facilitate the possible adoption of amplitude and phase shift keying (APSK) within the Aeronautical Mobile Telemetry (AMT) community, additional systems enhancements are necessary. The Coded APSK project has already demonstrated that the APSK waveform has the potential to yield significant gains in spectrum efficiency for AMT. These gains are due to the waveform's inherent spectrum efficiency in terms of occupied bandwidth and immunity to adjacent channel interference, which allows for greater spectrum packing density. However, given the nature of the AMT transmission channel, flight tests encounter both ground (multipath rich environment) and flight (highly dynamic environment) channel anomalies. To mitigate these anomalies, additional system enhancements such as (1) channel equalization, (2) space-time coding, and (3) diversity combining are required to aid APSK adoption. The successful development of the enhancements includes a prototype APSK transmitter, associated receiver/demodulator, followed by flight testing. These additional developments will usher along future adoption.